MUHAMMAD UMAR H

Trivandrum | +91 9188007536 | umarofcl10@gmail.com

WEBSITES, PORTFOLIOS, PROFILES

- **GitHub:** https://github.com/Mhd-Umar
- **Portfolio:** https://mhd-umar.github.io/Mhd-Umar_portfolio.github.io/
- **LinkedIn:** https://www.linkedin.com/in/muhammad-umar-h-90641034a/

PROFESSIONAL SUMMARY

Entry-level Software Developer and Computer Hardware Engineering diploma student (CGPA 9.71) skilled in Python, Django, C, Java, JavaScript, PHP, HTML, CSS, SQL and database management. Built a Django web application for college laboratory management with a team and is developing an AI and machine learning based ransomware detection system for Windows. Certified AI/ML Telecom Data Analyst (NSDC, Skill India). Seeking a Software Developer, Backend, Frontend, Hardware or IoT role.

SKILLS

- **Programming Languages:** Python, C, Java, JavaScript, PHP
- **Database:** MySQL, SQL, SQLite, Database Management (intermediate)
- **Networking:** Networking fundamentals (intermediate)
- **Hardware:** Computer hardware fundamentals
- **Frameworks and Tools:** Django, HTML, CSS, scikit-learn, VS Code
- **AI/ML and Data:** Data Science and Data Analysis, NumPy, Machine Learning (ML), Artificial Intelligence (AI) (intermediate)
- **Core Competencies:** Backend Development (intermediate), Frontend Development, Software Development, Cybersecurity Awareness, Problem-Solving, Hardware and IoT Fundamentals

WORK HISTORY

EDUCATION

Government Polytechnic College, Attingal, Kerala
Diploma, Computer Hardware Engineering (Lateral Entry), 2027
GPA: 9.71 (Semester 3 SGPA: 9.41, Semester 4 SGPA: 10.0)

Higher Secondary (Plus Two), 2024
GPA: 78.67%

Secondary School (SSLC / 10th), 2022
GPA: Full A+

CERTIFICATIONS

- AI/ML Telecom Data Analyst - Certified by NSDC (Skill India) through the Board of Vocational Higher Secondary Examinations, Kerala
- Hackathon 2026 - Participated up to district level in a hackathon by IEDC and Kerala Startup Mission

PROJECTS

ResoTrack – College Laboratory Resource Analyzer and Management System | 2026 | Python, Django, HTML, CSS, JavaScript, SQLite

- Developed in a team a Django web application with 3 role-based modules (student, lab administrator, main admin) to manage laboratory resources, stock, complaints and suggestions across 4 departments (CHE, EEE, AU, ME).
- Built a resource analyzer showing department-wise and lab-wise resources with working, damaged and under-maintenance status.
- Enabled lab administrators to add and update resources and stock (quantity, available, damaged, received) and generate stock and resource reports.

AI-Based Ransomware Early Detection and Warning System (Windows) | 2026 (In Progress) | Python, NumPy, Machine Learning, Isolation Forest

- Designing a behavior-based detector that identifies ransomware during encryption using 5 signals: file change speed, entropy jumps, extension changes, parent process and shadow copy deletion attempts.
- Implementing an AI risk score (0-100) using Isolation Forest and One-Class SVM, with SHAP explanations of every alert.
- Engineering an always-on-top alert with kill or suspend response for suspicious processes and a live dashboard. Runs fully on one PC.

PERSONAL INFORMATION

Title: Software Developer | Backend Developer | Frontend Developer | IoT and Hardware